National University of Computer and Emerging Sciences

Karachi Campus

Computer Networks(CS3001) Sessional-II Exam

Date: April 6th 2024

Course Instructor(s)

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Total Time (Hrs): 1
Total Marks: 30
Total Questions: 3

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Roll No Section Student Signature

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Attempt all the questions.

CLO # 1: Describe and evaluate the protocols, services and functions provided by each layer in the Internet protocol stack

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Q1: **[3+3+4 = 10 Marks]**

a) List down any three responsibilities of the rdt_send() i.e reliable data send function in reliable data transfer protocol.

b) Checksum is an error-detecting code used in many Internet standard protocols, including IP, TCP, and UDP. Following data blocks are received with Checksum= **0001111001101110**. You need to verify the integrity of the received data using the given checksum.

0111001010110011	**1011001110101000**	**1011101100110111**

c) A computer with IP Address: 192.28.33.21 sends a request to another computer. The request contains following UDP header: **0050CB84001C001C.**

Categorize content of UDP header by answering the following questions

i. What is the source port number?
ii. What is the socket address of the sender end?
iii. What is the total length of the user datagram?
iv. What is the length of data?

CLO # 2: Apply network protocol and communication services for client/server and other application layouts

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Q2: **[5+5 = 10 Marks]**

a) Let's suppose a window size of 4 and sequence number range from 0 to 8. Assume that 2nd packet in the window is lost. Apply TCP fast retransmission. Also, Generate Acknowledgements and complete the process until the lost packet is properly received by the receiver.

b) Let's suppose a window size of 4 and sequence number range from 0 to 8. Apply Go-Back-N (GBN) for sender and receiver events and actions. Assume that 3rd packet in the window is lost. Show the GBN operations using the diagram in the presence of the lost packet.

Q3: **[2+4+4= 10 Marks]**

a) Justify your answer.

i. Does TCP guarantees, meeting the delay requirement(s)? If yes, then discuss in brief, how such delay guarantee is provided by TCP?

ii. Is TCP segment of size 128KB is possible? If no, then what could be the maximum segment size (MSS) possible?

b) Using a simple telnet scenario, illustrate with the help of a timing sequence diagram, how the TCP handles sequence numbers and ACKs. The telnet client is Host A, and the server that is accessed remotely, is Host B. The initial sequence number 42 and 78 are selected by Host A and Host B respectively.

c) What action on the following events, will be taken by the TCP Receiver, for the ACK generation:

i. arrival of in-order segment with expected seq #. All data up to expected seq # already ACKed

ii. arrival of out-of-order segment higher-than-expect seq. # . Gap detected